

06-03-2026

## Assignment - 1

① Given initial no of Books = 120  
Perform the following operations:

Add 30 new books

Issue 25 books to students

Return 10 books

Remove 5 damaged books

Find the final no. of books in the library.

Program:-

```
Public class LibraryBooks {
```

```
    Public Static void main(String[] args) {
```

```
        int books = 120;
```

```
        books += 30;
```

```
        books -= 25;
```

```
        books += 10;
```

```
        books -= 5;
```

```
        System.out.println(books);
```

```
    }
```

```
}
```

O/P

130

2. Given Initial Score = 50

Perform the following operations:

Win Bonus +40 Points

Penalty -15 Points

Special Reward +30 Points

Time Penalty -10 Points

Find the final score.

Program:

```
Public class FinalScore {
```

```
    Public Static void main(String[] args) {
```

```
        int Score = 50;
```

```
        Score += 40;
```

```
        Score -= 15;
```

```
        Score += 30;
```

```
        Score -= 10;
```

```
        System.out.println(Score);
```

```
    }
```

```
}
```

Output

95

3. int a = 10;

int b = 6;

int c = 3;

int result1 = a & b | c;

int result2 = (a & b) & (b << 1);

result 1

a & b | c

$$10 \& 6 \rightarrow 1010 \& 0110 = 0010$$

$$= 2$$

2 | 3

$$0010 | 0011 = 0011 = 3$$

$$= 3$$

result 2

(a & b) & (b << 1)

(6 & 6) & (6 << 1)  $\rightarrow 1010 \& 0110 = 1100 = 12$

12 & (6 << 1)  $\rightarrow 1100 \& 1100 = 1100 = 12$

12 & 12

$$= 12$$

4. int a = 12;

int b = 5;

int c = 9;

int result1 = (a | b) & (c ^ a);

int result2 = (b << 2) | (c >> 1);

result 1

(a | b) & (c ^ a)

(12 | 5) & (9 ^ 12)  $\rightarrow 01100 | 00101 = 01101 = 13$

13 & (9 ^ 12)  $\rightarrow 01001 \& 01100 = 00101 = 5$

13 & 5  $\rightarrow 01101 \& 00101 = 00101 = 5$

$$= 5$$

result 2

(b << 2) | (c >> 1);

(5 << 2) | (9 >> 1)  $\rightarrow 0010 | 10100 = 10100 = 20$

20 | (9 >> 1)  $\rightarrow 01001 \>> 1 = 00100 = 4$

20 | 4  $\rightarrow 10100 | 00100 = 10100 = 20$

$$= 20$$